

[LLD] [Template] PRAJNA — Low-Level Design Document

[Module Name]

Version: 1.0

Date: [DD MMM YYYY]

Author: [Student Name]

Reviewer: [Mentor Name]

Status: Draft | In Review | Approved

Architecture Constraints (Read Before You Start)

All technology choices in your LLD **must** follow these project-wide constraints. These are non-negotiable unless you get explicit approval from the architect.

Language & Libraries

- TypeScript only — all libraries must be TypeScript-based or have TypeScript type definitions
- No exceptions without architect approval

Infrastructure

- AWS Serverless only — Lambda, API Gateway, DynamoDB, S3, Cognito, SQS, SNS, EventBridge, Step Functions, Aurora Serverless v2, Amazon Bedrock etc.
- No EC2, ECS, EKS, or self-managed servers
- Any exception requires written approval before including in your LLD

CDK

- All infrastructure must be defined using AWS CDK (TypeScript)
- Each module owns its own CDK stack

If you believe your module needs something outside these constraints, raise it in Section 13 (Open Questions) with your justification. Do not assume approval.

1. Module Overview

- What does this module do?
- Why does it exist in PRAJNA?

- Who are the end users of this module?
-

2. Scope

In Scope

- [List what this module covers]

Out of Scope

- [List what this module does NOT cover and which module handles it]
-

3. Dependencies

Depends On (modules this needs to work)

- [Module name] — [what it provides]

Depended By (modules that need this to work)

- [Module name] — [what it consumes]
-

4. Architecture & Design

Component Diagram

- [Describe or link a diagram showing how this module's components interact]

Data Flow

- [Step-by-step flow of data through this module — from input to output]
-

5. Data Model

Entities

- [List the key entities/tables this module owns]
- [For each: fields, types, constraints, relationships]

Access Patterns

- [How will this data be queried? List the key access patterns]
-

6. API Design

Endpoints

Method	Endpoint	Description	Auth Required	Request Body	Response

Error Handling

- [How does this module handle errors? What error codes/messages does it return?]

7. Technology Choices

For each choice, justify **why** you picked it over alternatives.

Area	Choice	Why	Alternatives Considered

8. Security Considerations

- [Authentication/authorization for this module]
- [Data privacy — who can see what?]
- [Input validation approach]
- [Any sensitive data handling?]

9. Testing Strategy

Type	What You'll Test	Tool/Approach
Unit		
Integration		
E2E		

10. CDK / Infrastructure

- [What AWS resources does this module need?]
- [CDK stack design — constructs, parameters, outputs]
- [How does this stack connect to other module stacks?]

11. Risks & Mitigation

Risk	Impact	Mitigation

12. Milestones & Timeline

Week	Deliverable
Week 1–2	LLD complete & approved
Week 3–4	
Week 5–6	
Week 7–8	
Week 9–10	
Week 11–12	

13. Open Questions

- [List anything you're unsure about that needs discussion with the team or mentor]

14. Self-Assessment — Amazon Leadership Principles

Review all 14 LPs below. Pick your **top 3 you demonstrated** and **top 3 you need to work on**. Provide an example for each.

#	Leadership Principle	Demonstrated / Need to Work On	Example from Your Work
1	Customer Obsession — Who is your module's customer? How did you design for them?		
2	Ownership — How did you take end-to-end ownership of your module?		
3	Invent and Simplify — Where did you simplify a complex problem?		
4	Are Right, A Lot — What decision did you make with incomplete information? How did it turn out?		
5	Learn and Be Curious — What new technology/concept did you learn for this module?		
6	Hire and Develop the Best — How did you help a teammate or learn from one?		

#	Leadership Principle	Demonstrated / Need to Work On	Example from Your Work
7	Insist on the Highest Standards — Where did you refuse to cut corners?		
8	Think Big — How does your module scale or support PRAJNA's long-term vision?		
9	Bias for Action — Where did you make a quick decision instead of over-analyzing?		
10	Frugality — How did you achieve more with less?		
11	Earn Trust — How did you handle disagreements or build trust with your team?		
12	Dive Deep — Where did you dig into the details to find the right solution?		
13	Have Backbone; Disagree and Commit — Did you push back on a decision? How?		
14	Deliver Results — Did you meet your milestones? What did you ship?		

Top 3 I Demonstrated:

- 1.
- 2.
- 3.

Top 3 I Need to Work On:

- 1.
- 2.
- 3.

15. References

- [Links to HLD, requirements doc, AWS docs, library docs, etc.]

PRAJNA — प्रज्ञा / Super-30